

R1 — ML Research Track, Week 1

Member: _____

Project setup (do first)

1. Fork the repo: <https://github.com/slvDev/esp32-ai>
2. Clone your fork. Add the original as upstream.
3. Read README.md, RESULTS.md, and research/tinystories/README.md.
4. Write 5 lines in the shared Drive doc: what you think the project does and where the hard part is.

Your task this week

Goal: the tokenized TinyStories data exists and works.

Steps: 1. Set up Python with uv. Run `uv sync` in the repo root. 2. Open Google Colab. Mount Google Drive. 3. Run the data prep: `uv run python -m research.tinystories.prepare --vocab 32768` 4. Move the data folder to Drive so week-2 runs reuse it: `data/tinystories/` 5. Open `data/tinystories/vocab-32768/tokenizer.json` and count the tokens.

Deliverable: a note in the tracking sheet saying data is ready, with the folder path and the token count.

Verify: the prepare command finishes without error. The tokenizer shows 32768 tokens.

Learning this week

Neural network basics. Watch episodes 1 to 3 of the 3Blue1Brown series. Write a small MLP in numpy that learns XOR. Run it and watch the loss fall.

Resources

- uv: <https://docs.astral.sh/uv/>
- Colab: <https://colab.research.google.com>
- 3Blue1Brown neural networks: https://www.youtube.com/playlist?list=PLZHQObOWTQDNU6R1_67000Dx_ZCJB-3pi
- TinyStories dataset: <https://huggingface.co/datasets/roneneldan/TinyStories>
- prepare.py source: <https://github.com/slvDev/esp32-ai/blob/main/research/tinystories/prepare.py>

Done when

- [] Data is prepared and cached in Drive
- [] Token count is recorded in the tracking sheet
- [] XOR MLP runs and the loss falls
- [] You can say what the project does in one sentence